

Jelizaveta Tsalaja

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EDUCATION

Bates College

Bachelor of Arts, GPA: 3.98/4.0, Majors: Mathematics, Digital & Computational Studies

Relevant Coursework: Mathematical Modeling (Python), Intro to Data Science (Visualization, R), Algorithms (Python)

Lewiston, ME

Sep 2023 - May 2027

London School of Economics and Political Science (LSE)

Visiting Student

Relevant Coursework: Data Science (Machine Learning, R), Advanced Data Manipulation (RAG, Python), Databases (SQL)

London, UK

Sep 2025 - Jun 2026

EXPERIENCE

IncQuery and UCLA Institute for Pure and Applied Mathematics (IPAM)

Industrial Research Intern & RIPS-NSF Scholar

Jun 2026 – Aug 2026

Los Angeles, CA

- Built an automated verification pipeline that formalizes all 42 operations in IncQuery's structured survey-logic language, enabling automated checks for unreachable questions and infeasible quotas across any IncQuery survey
- Co-authored the project's technical report and presented findings to 20 IncQuery engineers and 50+ attendees at IPAM's public research Project Day, translating technical work for both specialist and general audiences

Computer Vision Lab at Bates College

Research Fellow

May 2025 – Jul 2025

Lewiston, ME

- Conducted end-to-end implementation of image deblurring algorithms to improve point spread function estimation
- Extended blind image deconvolution pipeline from 2D to 3-dimensional kernel estimation for volumetric data, learning MATLAB to integrate with existing research codebase and implement 3D convolution operations

Analysis Group

Extern

Mar 2025

Boston, MA

- Quantified 60%+ combined market share in a mock Postmedia–Quebecor merger case using Excel, identifying antitrust risk and proposing regulatory intervention, while gaining exposure to the economic consulting industry

Bates College Digital & Computational Studies Department

Course-Attached Tutor (Intro to CS) & Drop-In Tutor (Academic Support Center)

May 2024 – Present

Lewiston, ME

- Deliver weekly group coding sessions for 50+ Intro to CS students and one-on-one/small-group tutoring in linear algebra, calculus, and data structures and algorithms, adapting instruction across formats and skill levels

PROJECT AND LEADERSHIP EXPERIENCE

Senior Thesis: Compartmental Modeling of Fare-Free Public Transit Policy

Aug 2026 – Present

- Exploring a compartmental modeling approach to study how fare elimination increase public transit ridership and distinguish ridership growth driven by new riders from increased travel frequency among existing riders over time

Text Mining: Corporate Communication During Economic Crises | LSE | [GitHub](#)

Oct 2025 – Jan 2026

- Analyzed 3,547 CEO departure announcements from S&P 1500 firms (2003–2014) using topic modeling, clustering, and keyness analysis in R to test whether corporate narratives shift during the 2008–2009 financial crisis
- Found crisis-period disclosures externalized blame to macroeconomic conditions, while normal-period disclosures emphasized individual agency, indicating firms shift messaging emphasis under economic stress

Influenza Spread Model

Oct 2024 – Jan 2025

- Utilized Python (NumPy, Matplotlib) to model the 2009 influenza outbreak at the U.S. Air Force Academy using compartmental modeling and ODEs, analyzing transmission rates, recovery periods, and intervention scenarios

Co-President, Bates College International Club

May 2024 – Present

- Manage \$3,000+ annual budget for 270+ member club, optimizing spending for high-impact cultural campus events
- Lead end-to-end planning for the International Dinner, coordinating registration and reimbursement for 35+ global cooking teams, and ticketing for 300+ attendees, showcasing campus cultural diversity

SKILLS

Languages: English, Russian (Fluent), Estonian (Fluent), French (Beginner)

Programming & Data Analysis: Python (NumPy, pandas), R, MATLAB, SQL, Excel, Git, LaTeX, Jupyter Notebooks

Interests: Ethics of data and AI, theater, public transportation, classical piano, hiking, dog training, home espresso